AUTOMATION / INVESTMENT ANALYSIS

THE AUTOMATION VALUE TRAP

How to read the investment decision graphic

ILLUSTRATIVE SCENARIO

20 reports/month | 3-year NPV | 10% discount rate

8 productive hours/month | EUR 30 incremental contribution/hour

01 / WHEN DOES AUTOMATION PAY OFF?

The upper panel maps 3-year net present value (NPV) against productive business demand and incremental cash contribution per hour. The white curve is the break-even boundary (NPV = EUR 0). Emerald indicates positive NPV; orange indicates negative NPV.

The selected point assumes 20 recurring monthly reports, 50 minutes of manual effort and 7 minutes of automated human effort per report. That releases 14.33 hours/month, but only 8 hours/month are assumed to have productive demand.

02 / WHY DOES THE VALUE CURVE FLATTEN?

The lower panel isolates the effect of released capacity while holding implementation and maintenance costs fixed. Once released capacity reaches 8 hours/month, productive demand is saturated. Further time savings do not generate additional cash benefit, so NPV remains flat.

If expanding automation requires additional setup or maintenance spending, NPV may decline after this point. The flat line is therefore a fixed-cost sensitivity view, not a prediction that scaling is free.

SCENARIO ECONOMICS

INITIAL INVESTMENT	ANNUAL MAINTENANCE	3-YEAR NPV
EUR 2,570	**EUR 730**	**EUR 2,777**

METHOD & EVIDENCE

Python execution time was benchmarked on synthetic retail data. Manual effort, human review time, productive demand, contribution per hour and financial costs are explicit scenario assumptions, not observed business cash flows. Released time is not counted as payroll savings.

PYTHON AUTOMATION / FINANCIAL MODELING / DECISION ANALYTICS